Spending IQ

From transaction data to financial intelligence — categorise, surface, act.

Revolut AI Challenge · HackDelft 2026
Executive Summary · 1M txns · ~17.8k users · Jul–Dec 2025

The problem. Revolut's shipped `category` column is a 1:1 copy of the raw MCC code — **396 granular labels** ("Record Shops", "Drinking Places") nobody budgets by — and MCC is **missing on 3.3%** of rows and often stale in production. Unusable categories → useless budgets → users who don't understand their money.

`cleaned txns → ① CATEGORISATION ENGINE (+category, +merchant_id, +is_recurring) → ② INSIGHT ENGINE (ranked insight objects) → ③ FINANCIAL AGENT — one data contract, three swappable engines`

① CATEGORISATION

A cheapest-confident-tier-first cascade onto a clean **12-category** budgeting taxonomy, designed to work **even when MCC is absent**.

- **Type rules** — FEE/ATM/credits (the 3.3% no-MCC rows)
- **MCC embedding** — description → nearest taxonomy anchor (cosine); the ~88% backbone
- **Name keywords** — Dutch head-words (Supermarkt, Tankstation) fill & cross-check
- **LLM tier (Claude)** — only the uncertain ~3%, batched & cached

Personalised: user-history features mean the *same* merchant lands in the category that fits *that* user (Albert Heijn £80 weekly = Groceries; £3 = a snack). Corrections feed back — the system learns your meaning.

② INSIGHT ENGINE

Pluggable detectors → severity ranking + dedup → the **one** insight that matters (no spam). Every number is computed, never modelled.

★ Decline Shield — preventive
"8 declines for insufficient balance, and Eetcafe Blauwe £16.55 is due in 3 days — top up."

★ Overspend Alert — personal baseline
"48% over your usual Digital this month" — vs your median, not a population average.

Supporting tiles: subscription radar, upcoming-charges forecast, peer benchmarking, fees. Detectors whose signal is weak in the data are **cut or capped, not faked**.

③ USER INTERACTION

The ranked hero becomes **one push**; the tap opens a Claude tool-calling agent.

- **Push** — "You could get declined again — tap to fix."
- **Explain** — agent answers with computed figures (decline history, next charge)
- **Plan** — finds the stretched category & the renewing sub
- **Act** — one-tap `set_budget` / `cancel_subscription`, gated by confirmation (simulated)

Model: `claude-opus-4-8`, adaptive thinking. The agent **plans & explains; it never invents a number** — every figure comes from a tool over the enriched table.

METHODOLOGY — USING SYNTHETIC DATA HONESTLY

The merchants are fictional, so **MCC *is* the ground truth** — we never claim to "beat" or "fix" it. We prove three real, verifiable things instead:

- **Normalisation** — collapse 396 unusable MCC labels into 12 budget-ready categories (the actual product value).
- **MCC-independent capability** — train on a time-ordered split (train Jul–Oct, test Nov–Dec) and predict the category on **held-out merchants with MCC withheld**: **47% accuracy vs a 25% majority baseline** — exactly the production case where MCC is missing.
- **Label-free checks** — coverage (we categorise the no-MCC rows), consistency, and personalisation lift.

Why these heroes: we validated every candidate on the full 1M rows and built only what's real. Decline Shield rides the **strongest signal in the data**; FX/fees & income forecasting were **cut** — effectively absent (≈£363 total FX markup; 4 users with recurring income).

SIGNAL IN THE DATA

Users hitting an insufficient-balance decline	77%
Declined transactions (75% insuff. balance)	19.2%
Spend in user's top-3 merchants (median)	67%
Cap-test accuracy, MCC withheld (vs 25%)	47%
Raw MCC labels → clean categories	396 → 12

Decline Shield turns the dataset's most universal pain into a **preventive** nudge — the difference between "here's where your money went" and "we stopped a problem before it happened."

~97%
VOLUME VIA RULES+EMBEDDING, ~0 MARGINAL COST

~3%
UNCERTAIN TAIL TO LLM — BATCHED & CACHED

1 push
RANKED HERO, NOT NOTIFICATION SPAM

0
HALLUCINATED NUMBERS — ALL COMPUTED

Impact: clean personal categories → trustworthy budgets → engagement; cutting insufficient-balance declines attacks failed-payment churn directly — a concrete Revolut win.